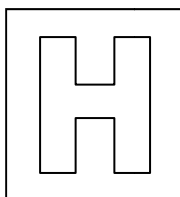


Candidate Name: _____

Class Adm No

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2014 Promotional Examination II Pre-university 2

H1 Biology

8875/01

22 September 2014

1 hour

Additional Materials: Optical answer sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, Adm No. and class on all the papers you hand in.

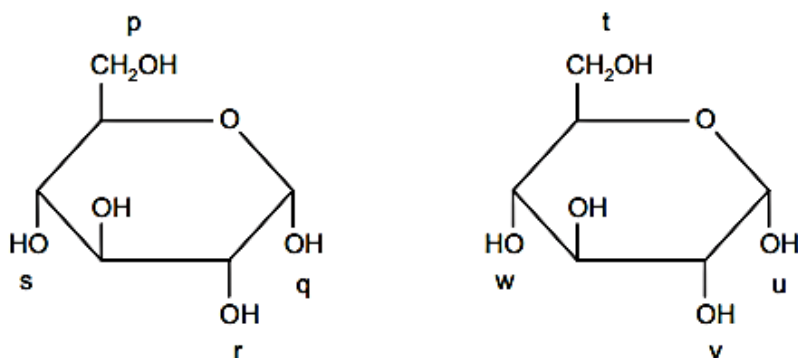
Do not use staples, paper clips, highlighters, glue or correction fluid.

Paper 1

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Multiple Choice Answer Sheet.

Calculators may be used.

1. The diagram shows two molecules of glucose. Four possible bonding positions are labelled p, q, r, and s, and t, u, v, w.



When these two molecules condense to form glycogen, where could the bonds be formed?

- A** p - u or p - v
B p - u or q - w
C p - v or q - w
D p - w or v - w
2. Which of the statements about polysaccharides can be used to describe both amylopectin and cellulose?

- I adjacent glucose molecules are rotated by 180°
 II contains 1,4 glycosidic bonds
 III polymer of α -glucose

- A** II only
B III only
C I & II
D I & III

3. One naturally occurring polysaccharide is an unbranched chain of the molecule acetylglucosamine linked by β -1, 4 glycosidic bonds. There are $-\text{CH}_2\text{OH}$ groups that alternate on each side of the polysaccharide chain.

Many hydrogen bonds form between these unbranched chains.

Which polysaccharide has a structure similar to that described?

- A Amylose
- B Amylopectin
- C Cellulose
- D Glycogen

4. What are the features of triglycerides?

	polar	less dense than water	higher energy value than carbohydrates	lower proportion of hydrogen than in carbohydrates
A	✓	✓	✗	✗
B	✓	✗	✓	✓
C	✗	✓	✓	✗
D	✗	✗	✗	✓

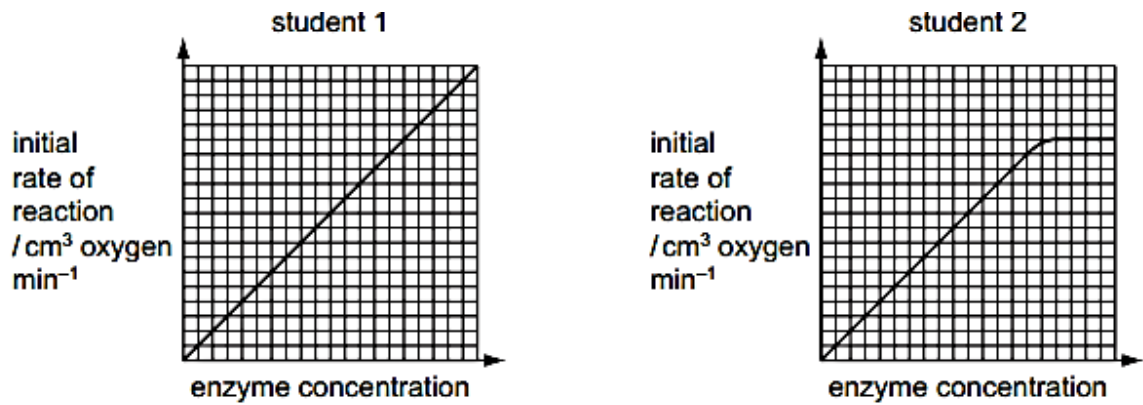
5. Features of four proteins, W, X, Y and Z, are described below.

- W consists of two identical polypeptides coming together to form a homodimer
- X consists of three polypeptides tightly coiled together
- Y consists of four polypeptides each with a prosthetic group
- Z consists of four polypeptides held together by disulfide bonds

Which row correctly identifies the named proteins?

	HIV Protease	Collagen	Haemoglobin
A	W	X	Y
B	W	Y	X
C	Z	X	Y
D	Z	Y	X

6. Catalase is an enzyme that catalyses the conversion of hydrogen peroxide into water and oxygen. Two students investigated the effect of enzyme concentration on the rate of reaction of the enzyme catalase. The students predicted their results would show the same trend. The graphs show the rates obtained by each student.

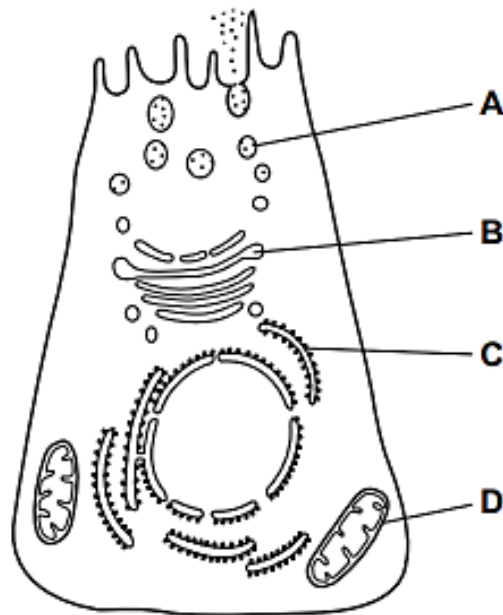


Which statement explains the different trend shown by student 2's results?

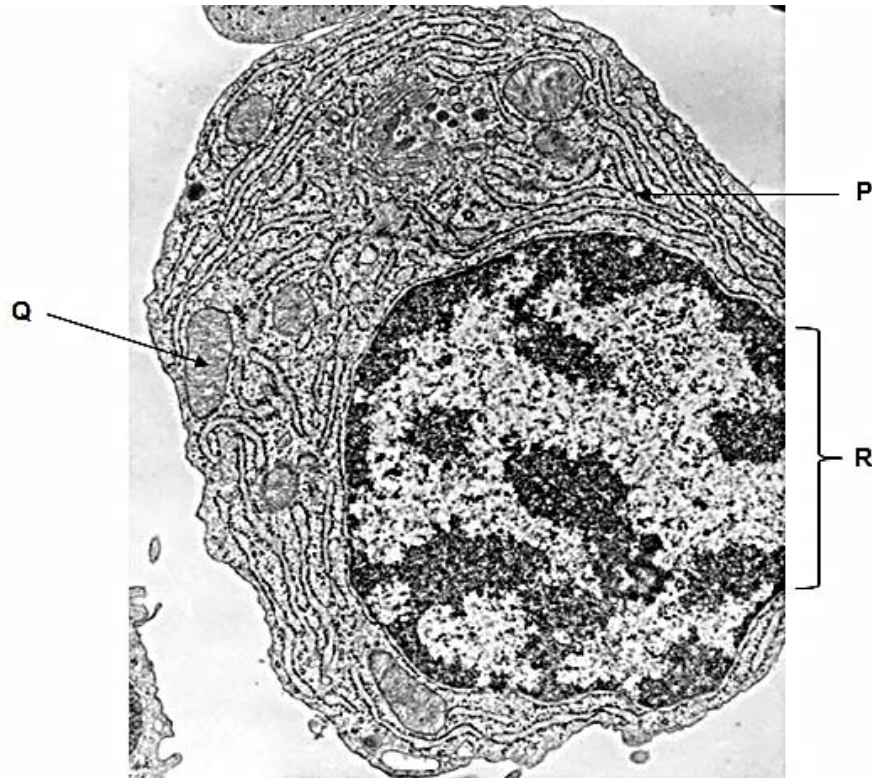
- A Student 2 included a competitive inhibitor in the investigation.
- B Student 2 performed the investigation at a higher temperature.
- C Student 2 performed the investigation at pH 6 compared to pH 8.
- D Student 2 used a lower concentration of substrate in the investigation.

7. The diagram is taken from an electron micrograph of a cell which secretes enzymes.

Where are these enzymes made?



8. The following diagram shows the electron micrograph of an actively secreting cell.



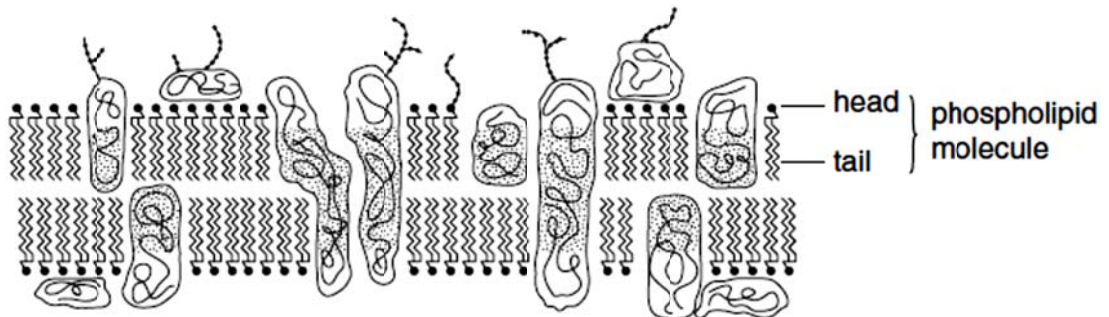
Which of the following about structures **P**, **Q** and **R** are correct?

	P	Q	R
A	Site of synthesis and transport of proteins	Consists of a double membrane	Product transported to P through vesicles
B	Increases surface area for attachment of ribosomes	Glucose is transported into Q for aerobic respiration to produce ATP	Consists of a double membrane
C	Involved in the synthesis and modification of proteins	Increases surface area for attachment of ribosomes	Contains both proteins and nucleic acids
D	Product transported to the Golgi bodies through vesicles	Transcription and translation occurs	Contains both DNA and RNA

9. Which row is correct for each of the components found on the cell surface membrane?

	Ion pump	Glycoprotein	Phospholipid	Cholesterol
A	A carrier protein involves in active transport	Has sugar chain attached to the protein	Contains 2 fatty acid chains and a phosphate head only	Restrains the phospholipids movements at high temperatures
B	Pumps ions against concentration gradient	Provides a receptor site for the signal ligand	Contributes to the fluidity of the membrane	Contains 4 interconnecting rings
C	Use ATP	Cannot move freely with the phospholipids	May be saturated or unsaturated fatty acid chains	Is a steroid
D	Has a hydrophilic channel to allow ions to diffuse through	For cell-to-cell recognition	Contains unsaturated hydrocarbon tails have kinks that allow the molecules to pack closely together	Is a lipid that does not contain fatty acids

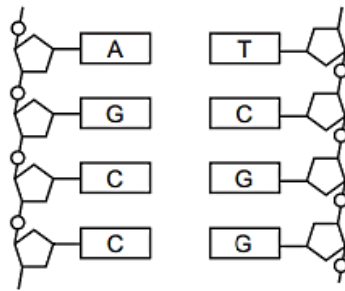
10. The diagram shows a section of a cell surface membrane.



What causes the phospholipid molecules to be arranged as shown?

- A** The heads are hydrophilic and point towards water molecules.
- B** The heads are hydrophobic and point away from water molecules.
- C** The tails are hydrophilic and point away from water molecules.
- D** The tails are hydrophobic and point towards water molecules.

11. The diagram shows part of a DNA molecule.



How many hydrogen bonds are involved in holding these strands of DNA together?

- A 11
 - B 9
 - C 8
 - D 4
12. DNA is said to replicate in a semi-conservative way. Results of Meselson and Stahl's experiments gave overwhelming support to this theory. They used *E. coli* which has a generation time of 50 minutes.

Here are the steps in their experiment but they are in the wrong order.

- I All bacteria contain ^{15}N DNA.
- II All bacteria contain hybrid DNA (^{15}N DNA and ^{14}N DNA).
- III Bacteria contain either all ^{14}N DNA or hybrid DNA.
- IV Bacteria grown in a ^{15}N medium for many generations.
- V Bacteria transferred to a ^{14}N medium and sampled every 50 minutes.

Which sequence of letters shows the correct order of the steps in the experiment?

- A I -> II -> III -> IV -> V
- B I -> IV -> V -> III -> II
- C IV -> I -> V -> II -> III
- D IV -> III -> II -> I -> V

13. The table below shows a list of characteristics displayed by mutant strains of *E.coli* during DNA replication and the possible reasons.

No	Characteristics	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is never completed	DNA ligase activity is missing
2	Supercoils are found to remain at the flanks of the replication bubbles	DNA topoisomerase is inactive
3	Synthesis of new strand is very slow	Substrate binding site of DNA polymerase is slightly altered
4	No initiation of replication occurs	A-T rich region at origin of replication is deleted

Which of the reasons correctly explain the characteristics displayed by the mutant bacteria?

- A 1 and 2
 B 1 and 3
 C 1, 3 and 4
 D All of the above
14. The table gives tRNA anticodons for four amino acids.

Amino Acid	tRNA anticodon
Asparagine	UUA
Glutamic acid	CUU
Proline	GGA
Threonine	UGG

A cell makes a polypeptide with the amino acid sequence:
 glutamic acid – asparagine – threonine – proline

What is the sequence of bases on the strand of the DNA that is complementary to the mRNA from which this polypeptide was formed?

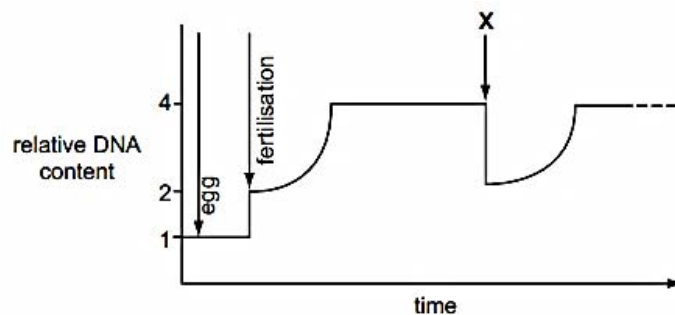
- A CTTTTATGGGGA
 B CUUUUAUGGGGA
 C GAAAATACCCCT
 D GAAAUAACCCCU

15. Translation did not occur successfully in a eukaryotic organism. Analysis of the structures in the cell revealed the following:

Structure	Presence/absence
Free floating aminoacyl-tRNAs	✓
Initiator tRNA- small ribosomal subunit complex	✓
Initiator tRNA (base-pairing with AUG codon) in the P-site of large ribosomal subunit	✓
Second aminoacyl-tRNA in the A-site of large ribosomal subunit	✓
Polypeptide chain	x

Which is the most probable explanation for these observations?

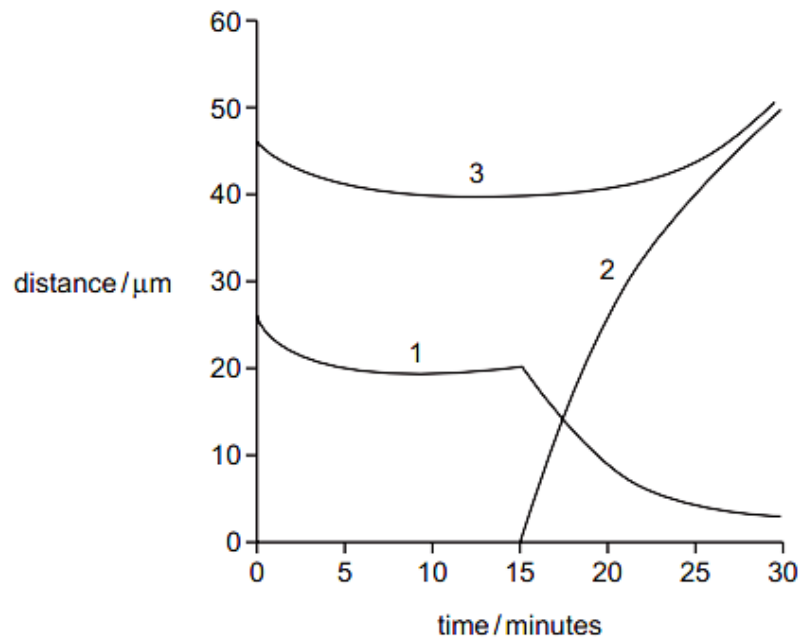
- A Aminoacyl tRNA synthetases fails to attach amino acids to tRNA.
 - B Peptidyl-transferase is not functioning properly.
 - C There is an error in the scanning for the start codon by ribosomal subunit.
 - D Release factor fails to hydrolyse polypeptide chain from the tRNA.
16. The graph represents the changes in the quantity of DNA present in one nucleus at different stages in the life cycle.



Which stage takes place at X?

- A interphase
- B metaphase
- C prophase
- D telophase

17. The graph shows three measurements obtained following metaphase of mitosis.



What measurements do the curves represent?

	distance between centromeres and poles of spindle	distance between centromeres of sister chromatids	distance between poles of spindle
A	1	2	3
B	1	3	2
C	3	1	2
D	3	2	1

18. Two types of plumage (feathers) exist in chickens: hen-feathering and cock-feathering. The type of plumage results from the expression of an autosomal gene locus that carries two alleles:

- Allele **H** (hen-feathering) is dominant in both female and male chickens.
- Allele **h** (cock-feathering) is recessive, and the phenotype is expressed only in male chickens.

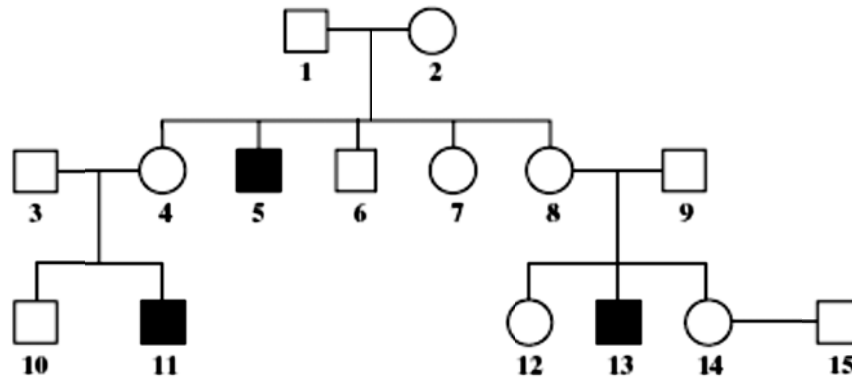
Which of the following would be the result of a mating between a cock-feathering male chicken, and a hen-feathering female chicken which is heterozygous at the gene locus?

- A** The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 50%.
 - B** The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 100%.
 - C** The probability of a male offspring being cock-feathering is 100% and that of a female offspring being hen-feathering is 50%.
 - D** The probability of a male offspring being cock-feathering is 100% and that of a female offspring being hen-feathering is 100%.
19. A population of cats contains individuals with shortened tails known as Manx cats and individuals with normal tails known as non-Manx cats. Mating of two non-Manx cats always produced non-Manx cats. Mating of two Manx cats always produced a mixture of Manx cats and non-Manx cats, where the number of Manx cats is usually twice that of non-Manx cats.

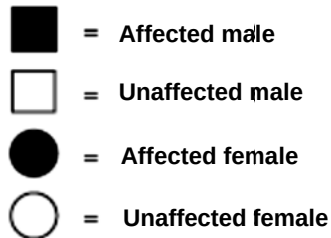
Which hypothesis is consistent with these observations?

- A** The genes for shortened tails and normal tails are linked on the same chromosome.
- B** The genes for shortened tails and normal tails are sex-linked.
- C** The allele for shortened tails is dominant over the allele for normal tails, but the homozygous dominant genotype is lethal.
- D** The allele for shortened tails is dominant over the allele for normal tails, but the homozygous recessive genotype is lethal.

20. The family tree shows the inheritance of a human genetic disorder. The genetic disorder causes degeneration of muscle tissue in humans.



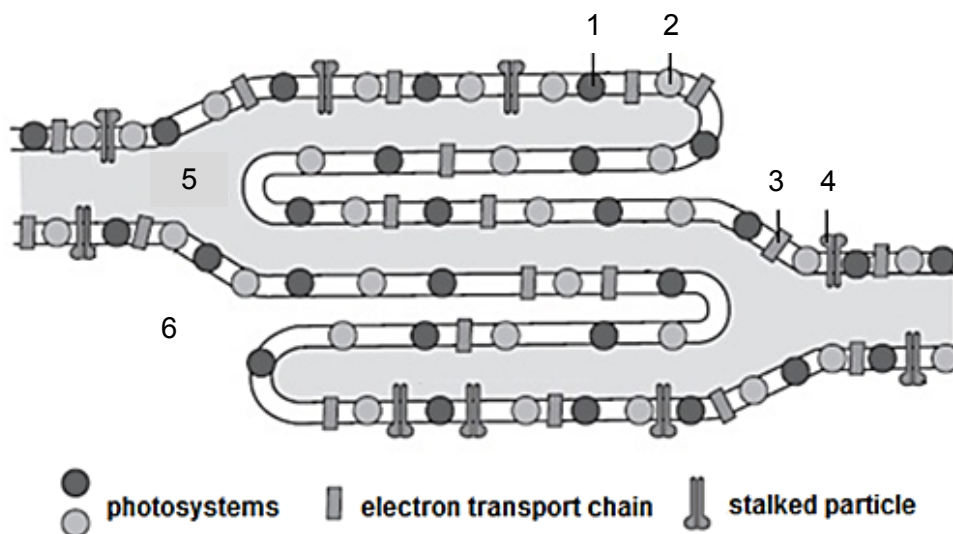
Key:



Which of the following statements can be concluded based on the diagram above?

- A** Individual 5 was affected with the genetic disorder as a result of genetic mutation during his lifetime.
- B** Individual 10 should be an affected male too, since his brother (individual 11) was affected with the genetic disorder.
- C** This genetic disorder is a sex-linked recessive trait.
- D** This genetic disorder is due to an autosomal dominant allele.

21. The figure shows the arrangement of photosystems, protein complexes containing chlorophyll molecules, on the thylakoid membrane of a plant chloroplast.



Which row correctly identifies the structure or function of the labelled regions?

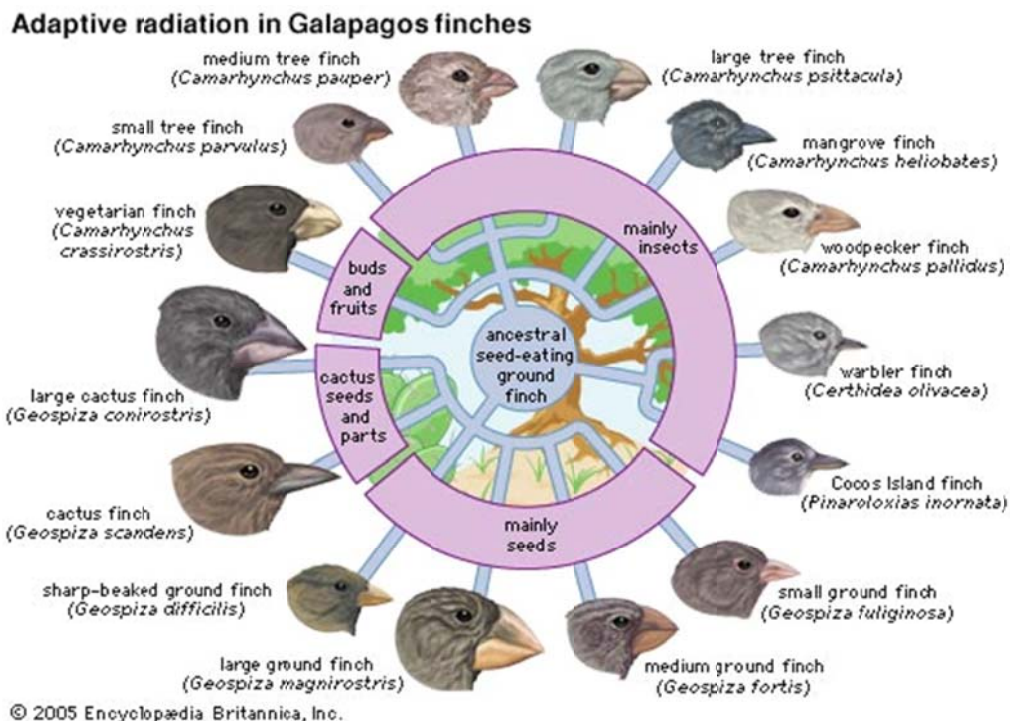
	site of photoactivation of chlorophyll	site of photolysis of water	site of ATP synthesis	stroma	thylakoid lumen
A	1 and 2	1 and 2	4	6	5
B	1 and 2	1	4	6	5
C	1, 2 and 3	1	3 and 4	5	6
D	1, 2 and 3	1 and 2	3 and 4	6	5

22. From yeast cells, a mixture of enzymes can be obtained that catalyses the conversion of glucose to carbon dioxide and ethanol.

Which combination of substances would increase the rate of ethanol production by these enzymes?

- A** ATP and lactate
- B** Reduced NAD^+ and coenzyme A
- C** NAD^+ and carbon dioxide
- D** ATP and reduced NAD^+

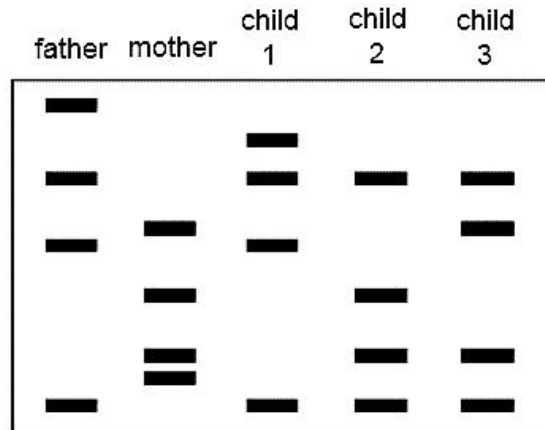
23. Which of the following describes the process of natural selection?
- A Different rates of reproductive success of different genotypes.
 - B Spontaneous occurrence of advantageous mutations.
 - C Change from simple to more complex organisms over time.
 - D Change in the size of population over time.
24. The Galapagos Islands are a group of volcanic islands in the eastern Pacific Ocean, about 1000km from South America. The diagram below illustrates the finches found on the islands; they resemble each other closely but differ in their feeding habits and in the shape of their beaks.



Which statement below explains how so many distinct and yet similar species arose.

- A As a result of the different food sources available, the ancestral species acquired different mutations which resulted in the emergence of a variety of beak shapes.
- B The finches were subjected to similar selection pressures which resulted in similar phenotypic characteristics in the distinct populations.
- C The difference in food sources served as a selection pressure, allowing only finches with certain beak shapes to survive and reproduce.
- D The finches migrated only to environments that they are already adapted to; finches with different phenotypes migrated to different islands.

25. DNA fingerprinting is often used to confirm the identity of an individual. The diagram shows a small section of the gel electrophoresis results from a DNA fingerprinting analysis of a family.



Which of the children is **least** likely to be the offspring of both parents?

- A Child 1
 - B Child 2
 - C Child 3
 - D All of the children are definitely offspring of both parents
26. Which of the following best describes the complete sequence of steps occurring during every cycle of PCR?
1. The primers hybridise to the target DNA.
 2. The mixture is heated to a high temperature to denature the double stranded DNA.
 3. Fresh *Taq* polymerase is added.
 4. *Taq* polymerase extends the primers to make a copy of the target DNA.

- A 2, 1 & 4
- B 3, 2 & 4
- C 2, 3, 1 & 4
- D 2, 1, 3 & 4

27. Which of the following statements is/are true regarding zygotic stem cells and cancer cells?

- I Both are able to move one from location to another.
- II Both are able to divide by mitotic division.
- III Both are specialised cells and capable to differentiate further.
- IV Both are capable of indefinite replication.

- A II only
- B I & III
- C II & IV
- D I, II & IV

28. A key feature of most multicellular organisms is the ability to differentiate and produce specialised cells.

Which row best describes the ability of zygotic cells to differentiate?

	totipotent	Pluripotent	multipotent
A	✓	✓	✓
B	✓	✗	✓
C	✓	✗	✗
D	✗	✓	✓

Key: ✓ = ability, ✗ = no ability

29. The genetically engineered super salmon was created from Atlantic salmon stocks and are capable of growing to a large size in 14 months.

Which of the following is not a benefit intended from the genetically engineered salmon?

- A They may not survive as well in the wild.
- B They are prone to further mutations.
- C They will produce sterile offspring with wild salmon.
- D They may replace wild salmon populations.

30. Which of the following could be a future application of the technology, skills, and data acquired from the Human Genome Project?

- I Individuals can have their DNA sequenced.
- II Researchers can determine genetic variations among different populations of humans.
- III Drugs can be developed based upon an individual's genetic makeup.

- A I only
- B II only
- C I & II
- D All of the above

End of Paper